Electromagnetic waves

Indroduction to EM. waves

Time varying induces Electric field in space J.C Maxwell analyzed that the converse is also true.

=> => Time varying Electric field induces Magnetic field in space

By various analysis Maxwell formulated a set of mathematical Equations involving electric field and magnetic field are called " Maxwell Equations "

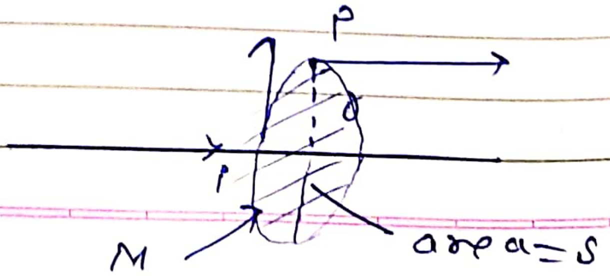
Basis of all laws of electromagnetism

Acc. to Maxwell equation in a region if EF or M.F varies with time then always These fields are coupled and vary in similar fasion & the em energy starts propagating called EM waves.

understanding Displacement current

Ampere's Law :
re-cap

for a st wire



By Ampere's Law we use

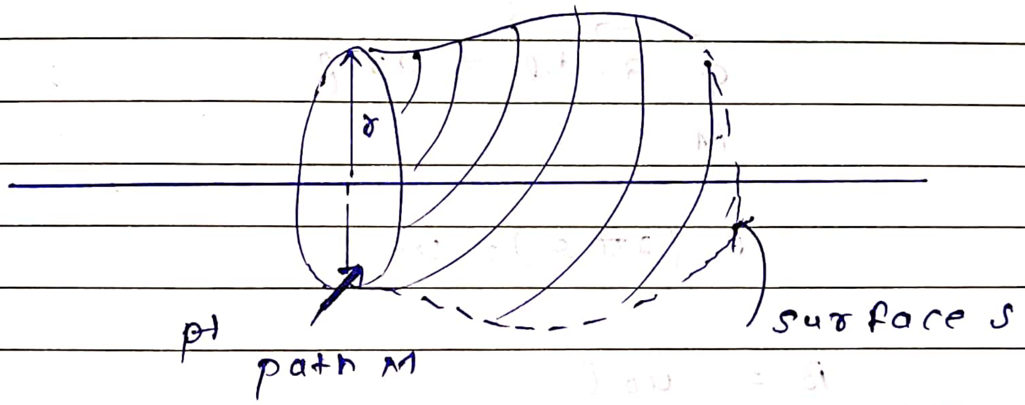
$$\oint_M \vec{B} \cdot \vec{dl} = \mu_0 i$$

current intercepted by the surfaces which spans the path M

$$B(2\pi r) = \mu_0 i$$

$$\Rightarrow \quad B = \mu_0 i / 2\pi r$$

surface S can take up any shape



$$\oint_M \vec{B} \cdot \vec{dl} = \mu_0 i$$

$$B(2\pi r) = \mu_0 i$$

$$B = \frac{\mu_0 i}{2\pi r}$$

Using Ampere's Law in surrounding of a charging capacitor.

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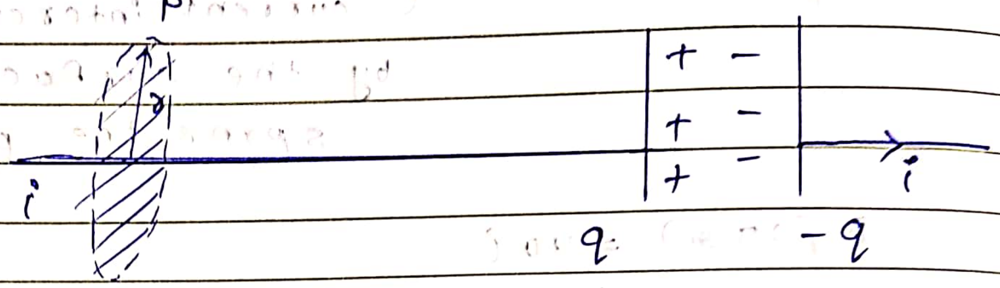

here $i = \frac{dq}{dt}$

By using Ampere's Law we have.

$$\oint_M \vec{B} \cdot \vec{dl} = \mu_0 i$$

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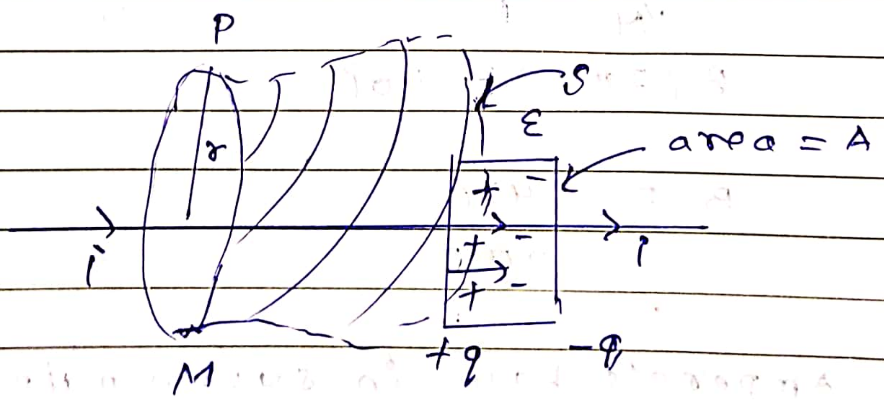
$$B = \frac{\mu_0 i}{2\pi r}$$

OR



The Electric field between capacitor is $E = \frac{q}{A \epsilon_0}$

Electric flux intercepting the surfaces is $\phi = EA = \frac{q}{A\varepsilon_0} \cdot A = \frac{q}{\varepsilon_0}$

Electric flux intercepting the surfaces is

$$\phi_E = EA = \frac{q}{A\,\varepsilon_0} \cdot A = \frac{q}{\varepsilon_0}$$

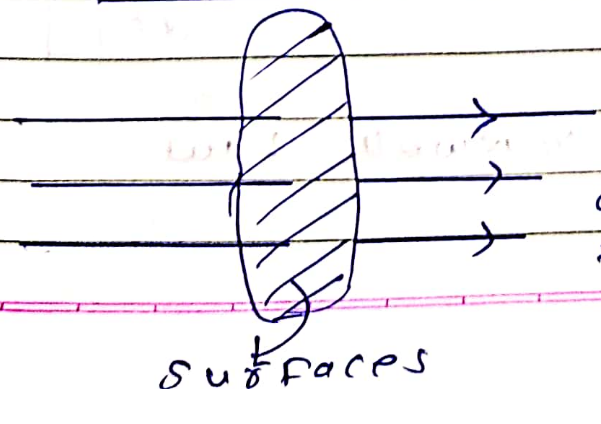
we have $\frac{d\phi_E}{dt} = \frac{1}{\varepsilon_0} \cdot \frac{dq}{dt} = \frac{1}{\varepsilon_0}\, i$

$\Rightarrow i = \varepsilon_0 \cdot \frac{d\phi_E}{dt}$ ← displacement current

Equivalent current intecepting the surface S, which is due to the change in electric flux $d\phi_E/dt$

Displacement current through a surface exist whenever there is a time varying Electric flux intercepted through the surface and this current magnitide is given as

$$\boxed{i_d = \varepsilon_0 \cdot \frac{d\phi_E}{dt}}$$



$E = f(t)$

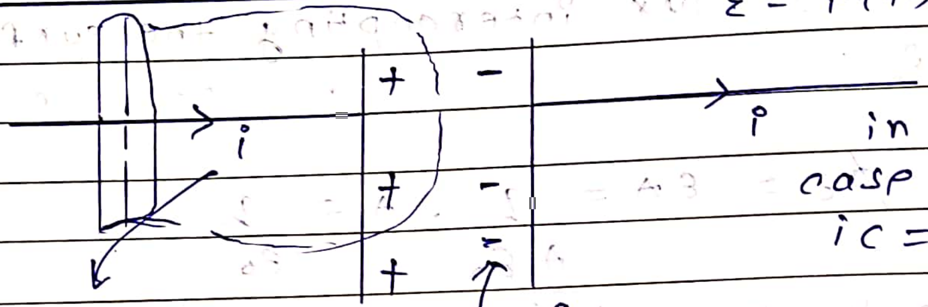
displacement current through S is

$i_d = \varepsilon_0 f'(t) \cdot S$

unit is Ampere (Amp) $\rightarrow$ A

Ampere – Maxwell's Law

$\mathcal{E} = f(t)$



i in this case $i_c = i_d = i$

conduction current

the region in which electric field is time varrying Displacement current is in existance

$$i_d = \varepsilon_0 \frac{d\phi_E}{dt}$$

In Ampere's Law Application for all cases we need to consider total current as

$$i = i_c + i_d = i_c + \varepsilon_0 \frac{d\phi_E}{dt}$$

Modified Ampere's Law

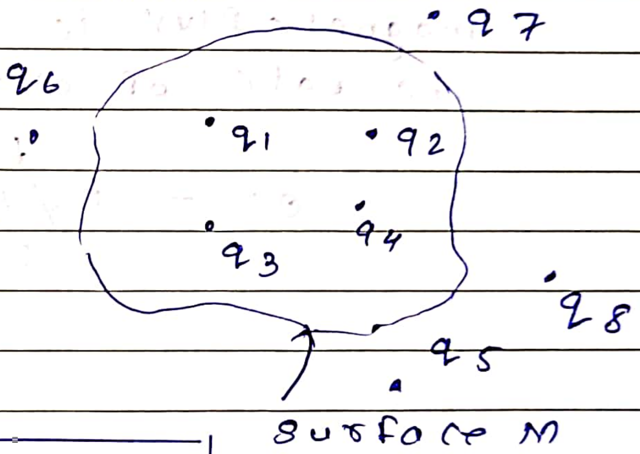
$$\boxed{\oint_M \vec{B} \cdot \vec{dl} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}}$$

Ampere - Maxwell Law.

Maxwell's Equation :

All the basic laws of electromagnetism can be formulated in four fundamental equations corresponding four basic laws, which are called maxwell's equation.

① Gauss's Law in Electrostatics

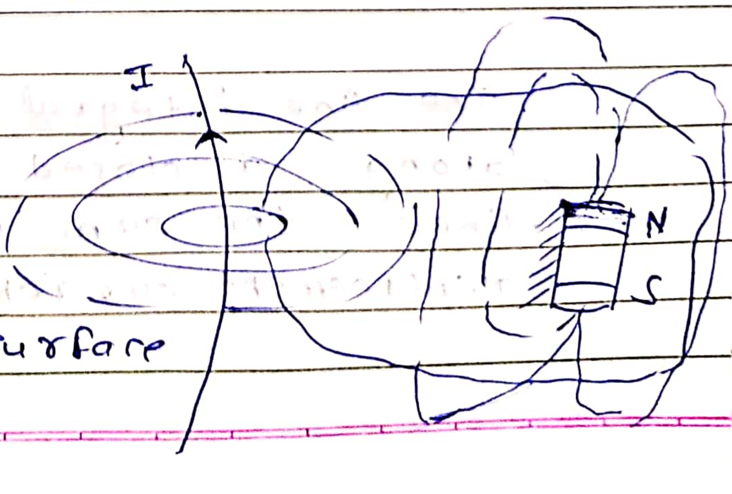


Total electric flux through any closed surface is equal to $\frac{1}{\varepsilon_0}$ times the net enclosed charge

$$\boxed{\oint_M = \oint_M \vec{E}\cdot\vec{ds} = \frac{q_{encl.}}{\varepsilon_0}}$$

② Gauss's Law in Magnetism:

"Total magnetic flux through a closed surface is equal to zero"



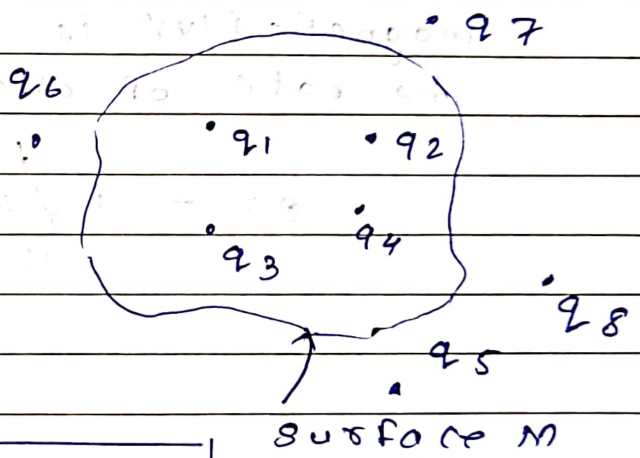
As a magnetic monopole does not exist, always magnetic flux through a close surface $= 0$.

Maxwell's Equation :

All the basic laws of electromagnetism can be formulated in four fundamental equations corresponding four basic laws, which are called maxwell's equation.

(1) Gauss's Law in Electrostatics

Total electric flux through any closed surface is equal to $\frac{1}{\varepsilon_0}$ times the net enclosed charge



$$\boxed{\oint_M = \oint_M \vec{E} \cdot d\vec{s} = \frac{q_{encl.}}{\varepsilon_0}}$$

(2) Gauss's Law in Magnetism:

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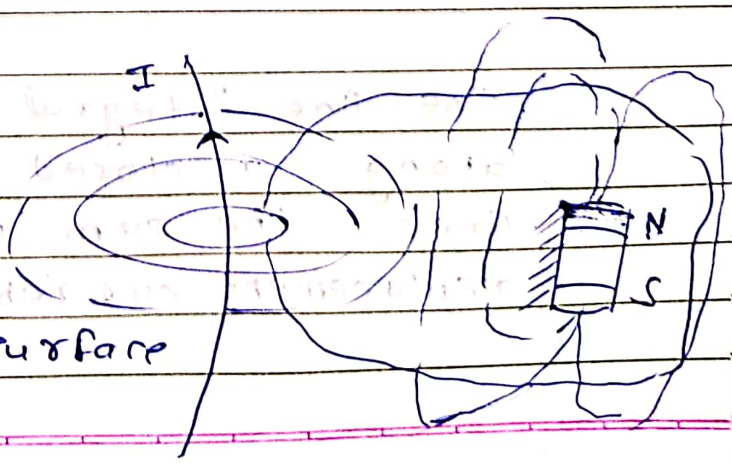
As a magnetic monopole does not exist, always magnetic flux through a close surface $= 0$.



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Because Magnetic lines are always closed loop lines

$$\boxed{\phi_M = \oint_M \vec{B} \cdot \vec{ds} = 0}$$

(3) Faraday's Law of E.M.I

"induced emf in a region of varying magnetic flux is equal to the line the rate of change of magnetic flux"

$$e = -\frac{d\phi_B}{dt}$$

-ve sign indicates that induced emf opposes the causes of change of flux.

For a given path we ~~cause~~ can use

$$\boxed{\oint_c \vec{E} \cdot \vec{dl} = -\frac{d}{dt} \oint \vec{B} \cdot \vec{ds}}$$

(4) Maxwell's Ampere's Law

The line integral of magnetic field along a closed path is equal to μ_0 lines the sum of conduction and displacement currents

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$$\oint \vec{B}\cdot\vec{dl} = \mu_0\left(I_c + I_d\right)$$

$$\boxed{\oint \vec{B}\cdot\vec{dl} = \mu_0\left(I_c + \varepsilon_0 \frac{d\phi_E}{dt}\right)} \quad (4)$$

Generation of EM waves

Maxwell's prediction--

— Time varying magnetic fields induces electric field, so vice verso should also be true

- in a region static and uniformly moving charges produce only one field

static charge ⟶ EF. is produced in surrounding

uniform current in a wire } → steady flow of moving charges

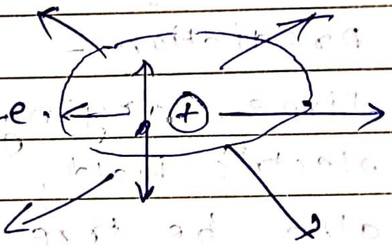
magnetic field produced in surrounding

- Maxwell's theory suggest an accelerating charge causes time varying electric field in surrounding and it induces magnetic field in the region and due to time ~~varying~~ variation of magnetic field again an Electric field is induced and so on.

Due to continious variation in Electric field and magnetic field in a region electromagnetic energy starts propogating which are called E-M waves

$\boxed{\text{Source of EM - waves}}$ $\longrightarrow$ $\boxed{\text{accelerating charge.}}$

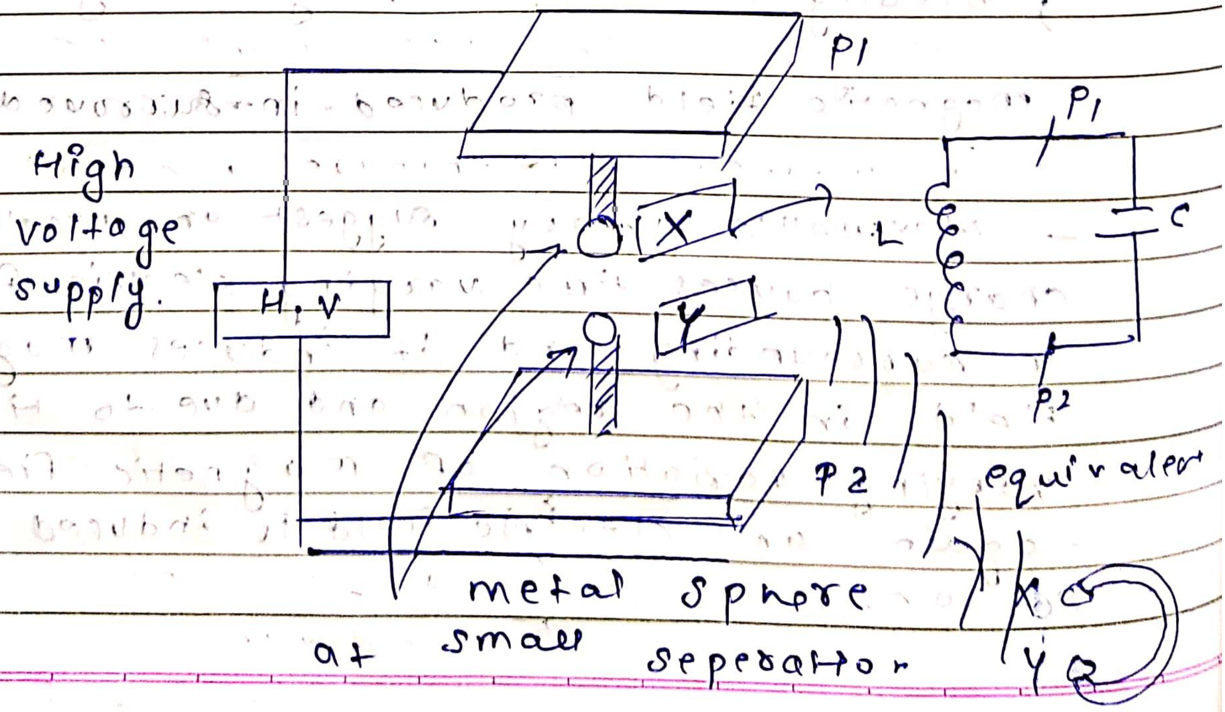
oscillating charge.



in 3-D space, an oscillating charge starts radiating E-M waves.

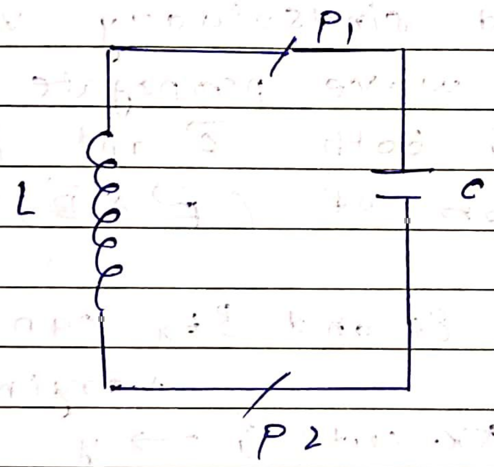
$\boxed{\text{Hertz Experiment}}$ (1887)

First experiment demonstrated of EM waves radiation and detection



$P_1 P_2$ behave like a capacitor and due to voltage low seperation between spheres x and Y. The high Electric field between sphere causes dielectric breakdown of air and thus in ionized sphere elelectric ~~breakdown of air~~ spark is produced

The inductance between plates is v. low as all rods and ~~cond~~ connecting wires are conducting.



Due to low values of L and C, the oscillation frequency in v. high given as

$$f = \frac{1}{2\pi\sqrt{LC}}$$ (here it is $\sim 10^7$ Hz)

Due to high frequency variation in spark of $\vec{E}$ and $\vec{B}$ of oscillating charges ~~Electr~~ EM waves are radiated in surrounding. if the gap in ~~decter~~ dector g coil is parallel to spark gap E M waves can be detected.

Nature of EM-waves

Maxwell gave the idea of EM waves and Hertz, J C-Bose and other physist experimentally demonstrated these waves based on their experiments and study. About EM waves it was analysed and accoplied that:-

1) EM waves are transverse in nature with $\vec{E}$ and $\vec{B}$ perpendicular to each other and sinusoidally varying with time and wave propogate in direction normal both $\vec{E}$ and $\vec{B}$ or along the direction of $(\vec{E} \times \vec{B})$

The functions of E_t and B_{tx} can be given as

varying along

$$E = E_0 \sin(kx - \omega t)\,\hat{j} \rightarrow y$$
$$B = B_0 \sin(kx - \omega t)\,\hat{k} \rightarrow z$$

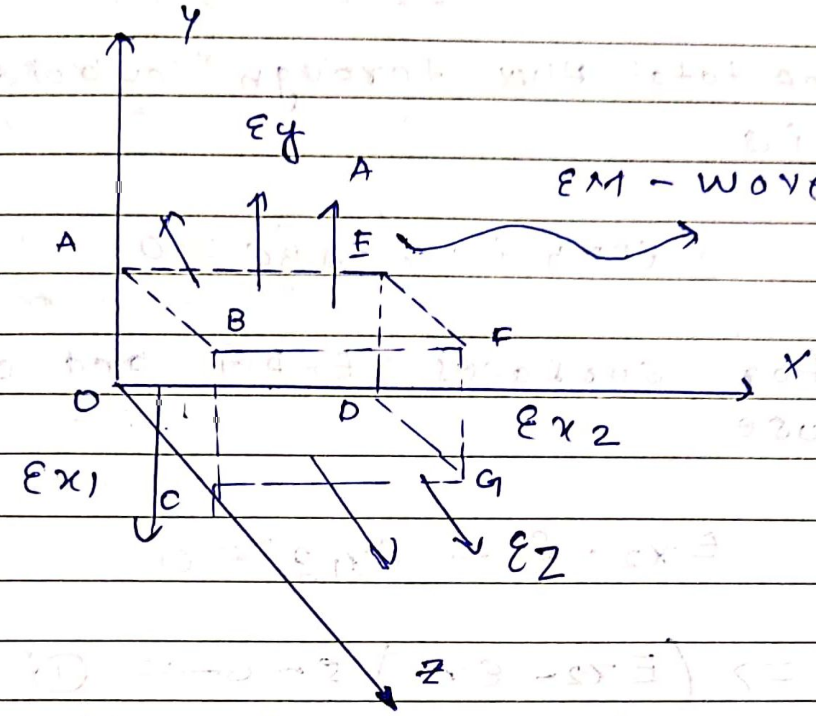
(2) The relation in amplitudes of electric fields and magnetic fields in EM waves is

$$E_0 = cB_0$$

(3) The speed of Electromagnetic waves in free space is equal to $\frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ is c

That is speed of light
3×10^8 m/s

Transverse Nature of EM-waves



if we consider Electromagnetic wave is travelling in x direction => $\vec{E}$ is varying with x and t

. if Ex, Ey, and Ez are components of $\vec{E}$ then for a cuboid surface as shown in figure, we have from

Gauss's Law

$\phi_{cuboid} = 0$ as no net charge is enclosed in it.

As Ey and Ez are not changing with y and z so we can say that in y direction

$\phi_{ABEF} = -\phi_{ODGC}$ | in z dir

$\phi_{BFGC} = -\phi_{OAED}$

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The total flux through cuboidal surface is

$$\phi_{EFDG} + \phi_{OABC} = 0$$

for surfaces EFDG and OABC we use

$$E_{x_2} \cdot S - E_{x_1} S = 0$$

$$\Rightarrow (E_{x_2} - E_{x_1}) S = 0 \quad (1)$$

from eq. (1) we have

Either $E_{x_2} - E_{x_1} = 0 \Rightarrow E_{x_2} = E_{x_1}$

OR $E_{x_2} = E_{x_1} = 0$ not possible

we have as it is

$E_{x_2} = E_{x_1} = 0$ not static field

Thus in above analysis, it is clear that along the direction of propagation of EM wave, Ef does not have any component

$\Rightarrow$ Ef is oscillating in direction $\perp$ to direction of propagation.

similarly we can state this for magnetic field, Hence EM waves are transverse in Nature

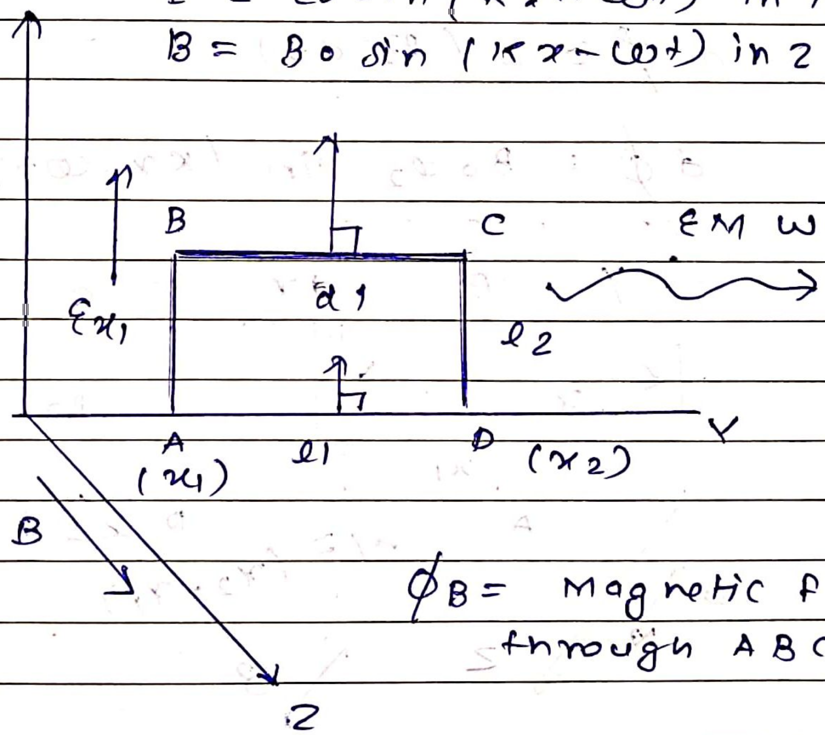
Relation in Amplitudes of EF and MF in EM wave.

if in an EM wave travelling in x-direction $E = f(x,t)$

and $B = f(x,t)$ are given as :-

$E = E_0 \sin(kx - \omega t)$ in y direction

$B = B_0 \sin(kx - \omega t)$ in z direction



ϕ_B = Magnetic flux through ABCD

By Faraday's Law of Electromagnetic Induction in loop ABCD

we can use $\displaystyle\oint_{ABCD} \vec{E}\cdot\vec{dl} = -\frac{d\phi_B}{dt}$

here $$\oint_{ABCD} \vec{E}\cdot\vec{dl} = \int_A^B \vec{E}\cdot\vec{dl} + \int_B^C \cancel{\vec{E}\cdot\vec{dl}} + \int_C^D \vec{E}\cdot\vec{dl}$$

$$+ \int \cancel{\vec{E}\cdot\vec{dl}}$$

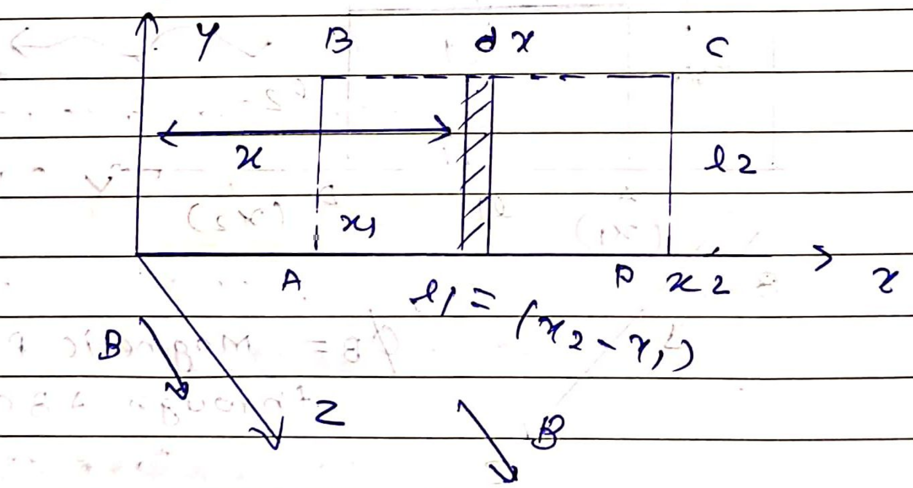
$$= E_{x_1} l_2 - E_{x_2} \cdot l_2 = E_0 l \left[\sin(kx_1 - \omega t) - \sin(kx_2 - \omega t) \right] \quad \text{--- (1)}$$

Relation in amplitude of Electric and Magnetic field.

The magnetic flux through the elemental strip shown at an instant is

$$d\phi = B(l_2\, dx)$$

$$d\phi = B_0 l_2 \sin(kx - \omega t)\, dx$$



Total magnitude flux through loop ABCD is

$$\phi_{AB} = \int d\phi = \int_{x_1}^{x_2} B_0 l_2 \sin(kx - \omega t)\, dx$$

$$= -\frac{B_0 l_2}{k} \left[\cos(kx - \omega t) \right]_{x_1}^{x_2}$$

$$= -\frac{B_0 l_2}{2}\left[\cos(kx_2 - \omega t) - \cos((kx_1) - \omega t)\right]$$

RHU → $$\frac{d\phi_B}{dt} = -\frac{B_0 l_2 \omega}{K}\left[\sin(kx_2 - \omega t) - \sin(kx_1 - \omega t)\right]$$

from eqn ① and ② we have in magnitudes $\varepsilon_0 l_2 = B_0 l_2 C$

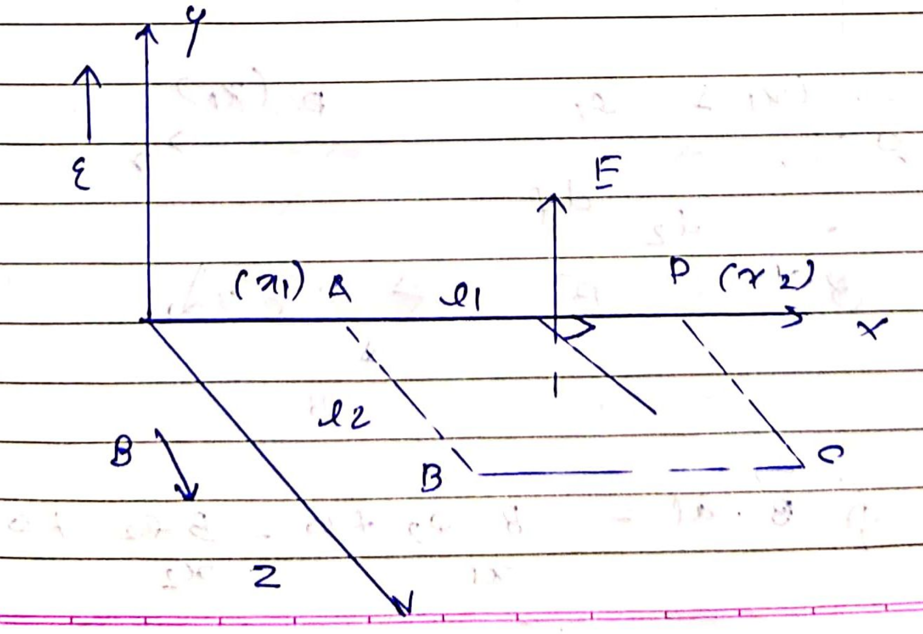
$$\Rightarrow \varepsilon_0 = B_0 C$$

$\boxed{\text{speed of EM waves.}}$

if E and B are electric field and magnetic field in EM waves travelling along x-direction and their variation functions are given as

$E = \varepsilon_0 \sin(kx - \omega t)$ (in y direction)

$B = B_0 \sin(kx - \omega t)$ (in z direction)



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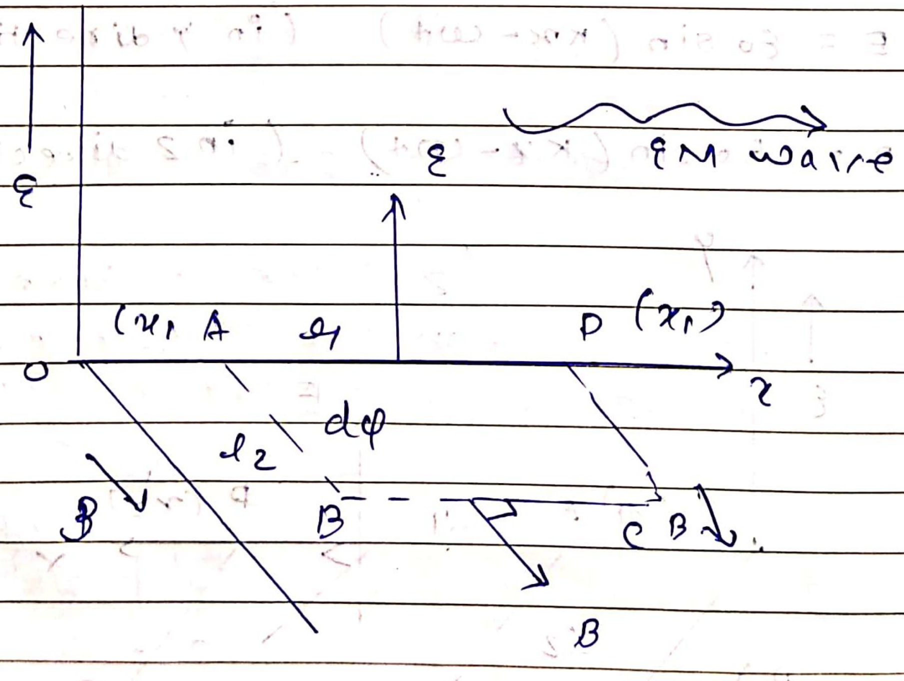
if we use Ampere-Maxwell's Law on loop ABCD, we have

$$\oint_{ABCD} \vec{B} \cdot \vec{dl} = \mu_0 \left(\epsilon_0 \frac{d\phi_E}{dt} \right)$$

[As condition current through ABCD is zero)

$$\oint_{ABCD} \vec{B} \cdot \vec{dl} = \int_A^B \vec{B} \cdot \vec{dl} + \int_B^C \vec{B} \cdot \vec{dl} + \int_C^D \vec{B} \cdot \vec{dl} + \int_A^B \vec{B} \cdot \vec{dl}$$

$$\Rightarrow \oint \vec{B} \cdot \vec{dl} = B l_2 + 0$$



$$= \oint \vec{B} \cdot \vec{dl} = \underset{x_1}{B} l_2 + 0 - \underset{x_2}{B} l_2 + 0$$

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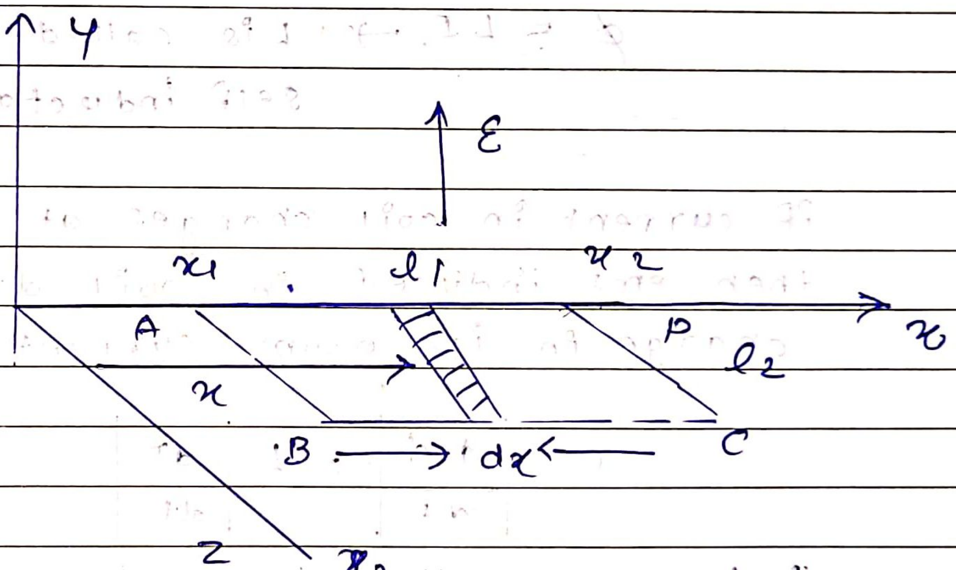
$$\oint_{ABCD} \vec{B}\cdot\vec{dl} = (Bx_1 - Bx_2)\,l_2$$
$$= B_0 l_2 \left[\sin(kx_1-\omega t) - \sin(kx_2-\omega t)\right]$$

Electric flux through loop ABCD is given as

flux through elemental strip is

$$d\phi = E\cdot(l_2\,dx)$$
$$d\phi = E_0 l_2 \sin(kx-\omega t)\,dx$$



$$\phi_E = \int d\phi = E_0 l_2 \int_{x_1}^{x_2} \sin(kx-\omega t)\,dx$$

$$= \frac{E_0 l_2}{k}\left[\cos(kx_2-\omega t) - \cos(kx_1-\omega t)\right]$$

$\boxed{c = \omega/k}$

$$RHS = \mu_0 \varepsilon_0 \frac{d\phi}{dt} = \mu_0\varepsilon_0\left(\frac{E_0 l_2 \omega}{k}\right)\left[\sin(kx_2-\omega t) - \sin(kx_1-\omega t)\right] \quad (3)$$

From eq - (2) and (3) we get LHS = RHS

$$\mu_0 \varepsilon_0 E_0 c = B_0 \Rightarrow \mu_0 \varepsilon_0 \cancel{B_0} c^2 = \cancel{B_0}$$

$$c^2 = \frac{1}{\mu_0 \varepsilon_0} \rightarrow c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3\times 10^8 \text{ m/s}$$

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